

Sridhar Charan Chekka

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Professional Summary

AI/ML Engineer specializing in **Generative AI** and intelligent systems, with deep expertise in **Retrieval-Augmented Generation (RAG)**, **Multi-Modal RAG (MRAG)**, and **agentic AI architectures**. Proven experience designing and deploying scalable, production-grade AI solutions that integrate **multi-agent workflows**, **deep learning**, and **real-time data pipelines** for advanced anomaly detection, contextual reasoning, and operational intelligence. Focused on building autonomous, decision-capable systems that leverage structured and unstructured data to deliver high-impact, data-driven insights across complex, real-world environments.

Skills

AI Applications: Real time Monitoring Systems, **Anomaly detection**, Predictive Analysis, Intelligent Decision Support Systems, **Multimodal AI Systems**, AI Automation & Agentic Workflows

Core AI Technologies: Machine Learning, Deep Learning, Generative AI, Time Series Forecasting, Computer Vision, NLP, **RAG &MRAG**, **Agentic AI**

Languages: Python, MySQL

Frameworks: PyTorch, TensorFlow, **Scikit Learn**, Hugging Face, **LangChain**, **LangGraph**, CrewAI, Autogen, AgentScope

ML Libraries: NumPy, Pandas, NLTK, Spacy, Streamlit, Flask, FastAPI

Tools & Platforms: Git, Jupyter Notebook, Google Colab, VS code, Vector Databases, Docker, Power BI, N8N(Basic), Claude code, Codex

Cloud Platforms: AWS(Basic)

Accomplishments

- District-Level Dhirubhai Ambani Quiz Runner-Up – Demonstrated broad subject knowledge and competitive spirit.
- State-Level Elocution Contest Runner-Up – Showcased public speaking skills and persuasive communication.

Experience

Machine Learning Engineer, V-OptimAIse Oct 2025 – Present

- Designed and deployed advanced AI solutions for **industrial applications**, including statistical machine learning, time-series forecasting, Transformer models, Computer Vision systems, and **Retrieval-Augmented Generation (RAG)** architectures to enable real-time operational intelligence.
- Led development of **FurnaceMind**, an AI-powered blast furnace intelligence platform that performs real-time monitoring, anomaly detection, and automated decision recommendations using **multi-agent** autonomous workflows and Generative AI technologies.

AI Research Intern, CortexTor Labs June 2025 – Sep 2025

- Conduct cutting-edge research in multimodal AI for education, specializing in **multimodal RAG systems** and **AI agents** integrating text, image, and audio modalities.
- Contribute to peer-reviewed publications and develop open-source tools advancing AI-driven educational solutions.

AI/NLP Intern, ExcelR Dec 2024 – April 2025

- Built a context-aware text summarizer using transformer models like BERT and T5 to generate accurate and concise document summaries.
- Fine-tuned models on domain-specific data, improving summary coherence and boosting ROUGE scores.

ML Intern, Edunet Foundation June 2024 – July 2024

- Predicted employee burnout with **96%** accuracy by implementing a Linear Regression model on real-world HR data, which led to improved insights for proactive workforce management.
- Streamlined data workflows by performing preprocessing and feature engineering using Python and Scikit-Learn, which led to a more robust and interpretable prediction pipeline.

Projects

Symp_bot – AI Doctor Assistant Link

- Developed a multimodal AI assistant that analyzes voice and image inputs to provide symptom analysis and disease suggestions using the Llama-4 Maverick LLM, enabling intelligent conversational healthcare support.
- Built an end-to-end application integrating Generative AI, voice interaction (ElevenLabs), and a Gradio interface, demonstrating real-time AI reasoning and accessible user interaction.

- Tools Used: Python, Gradio, Multimodal AI, Git, GitHub, VS Code

AI-Powered Travel Assistant – Journey Bot

Link

- Built a multi-agent travel planner using Crew AI, LangChain, and Llama-4-Scout-17B for personalized itinerary generation.
- Designed a Streamlit interface for interactive trip planning.
- Tools Used: Python, CrewAI, LangChain, Streamlit, Git, GitHub, VS Code

Source Code Analyzer

Link

- Developed a tool to fetch Git repositories and provide code explanations based on user queries.
- Integrated LangChain, RAG, and Qwen/Qwen3-4B for enhanced contextual accuracy and used Flask for frontend.
- Tools Used: Python, Flask, LangChain, **RAG**, Chroma DB, Hugging Face, GitPython, Jupyter Notebook

Certifications

- Machine learning with Python: A Practical Introduction – IBM (2023)
- Google Analytics Certification - (2025)
- MLOps: Real-World Machine Learning Projects for Professional – Udemy (2025)
- AI & Deep learning, Natural Language Processing – ExcelR (2025)

Education

Bachelor of Techonology	Sept 2022 – Apr 2025
Electrical and Electronic Engineering	CGPA: 7.8
Diploma	June 2019 – July 2022
Electrical and Electronic Engineering	CGPA: 8.03
SSC	April 2019
	CGPA: 10.00